

Solutions – Practice Problem Set 4

ECON 101 – Macroeconomics

Fall 2025

1. **Solution.** In standard growth accounting, long-run growth in real GDP stems from three sources:

- *Capital deepening* (more physical capital per worker): Expands the productive capacity per worker but is subject to diminishing marginal returns. **Policy example (Canada):** Accelerated capital cost allowance (ACCA) or investment tax credits to lower the user cost of capital and stimulate private investment.
- *Labour expansion and quality* (hours worked and human capital): More workers/hours raise aggregate output; higher education and skills raise output per worker. **Policy example:** Targeted immigration of high-skill workers; subsidized apprenticeships and upskilling programs.
- *Total Factor Productivity (TFP)* (technology and efficiency): Better ideas, organization, and diffusion increase output for given inputs. **Policy example:** R&D subsidies and stronger IP enforcement to stimulate innovation and diffusion.

Interactions matter: education and R&D raise the return to capital; institutions (property rights, competition policy) support TFP growth.

2. *Given:* $Y = AK^{0.3}L^{0.7}$, with $A = 5$, $K = 400$, $L = 100$.

(a) **Level of real GDP.**

$$Y = 5 \times 400^{0.3} \times 100^{0.7} \approx 5 \times 5.757 \times 26.334 \approx \mathbf{757.86}.$$

(Any answer ≈ 758 earns full credit.)

(b) **Percentage increase in Y if K rises by 10% (holding L fixed).**

With a Cobb–Douglas, the elasticity of Y with respect to K equals the capital exponent 0.3:

$$\% \Delta Y \approx 0.3 \times 10\% = \mathbf{3.0\%}.$$

Using the exact change: $(1.1)^{0.3} - 1 \approx \mathbf{2.90\%}$.

(c) **Why $< 10\%$?**

Diminishing marginal returns to capital: holding L and A fixed, each additional unit of K raises output by less than the previous one. Hence, a 10% increase in K yields only about a 3% increase in Y .

3. *Given:* $g_Y = 4\%$, $g_K = 2\%$, $g_L = 1\%$, factor shares $\alpha_K = 0.4$, $\alpha_L = 0.6$. The decomposition:

$$g_Y \approx \alpha_K g_K + \alpha_L g_L + g_A.$$

(a) **Contributions.**

$$\text{Capital: } 0.4 \times 2\% = \mathbf{0.8\%}, \quad \text{Labour: } 0.6 \times 1\% = \mathbf{0.6\%},$$

$$\text{TFP: } g_A = 4\% - (0.8\% + 0.6\%) = \mathbf{2.6\%}.$$

(b) **Interpretation of negative TFP.**

Negative TFP growth implies that, for given inputs, the economy becomes *less* efficient—e.g., due to misallocation, regulatory frictions, supply disruptions, or slower diffusion of technology. Output grows more slowly than input growth would predict.

4. Solution.

- (a) *R&D subsidies*: Raise private returns to innovation $\Rightarrow$ faster idea creation/diffusion $\Rightarrow$ higher TFP & long-run per-capita GDP.
- (b) *Higher education*: Expands human capital, complements technology and capital, and raises the speed of adoption $\Rightarrow$ higher labour productivity and TFP.
- (c) *Property rights/anti-corruption*: Improves contract enforcement and investment security, reduces rent-seeking, allocates resources to high-productivity uses $\Rightarrow$ higher TFP and capital formation.

5. Solution. In a closed economy, $AE = C + I + G$; in an open economy, $AE = C + I + G + NX$.

In Canada and other advanced economies, *consumption* is typically the largest component.

Planned investment (I^P) is firms' intended spending on structures, equipment, and inventories; *actual* investment equals I^P plus *unplanned* inventory changes that occur when sales differ from expectations.

6. Given: $C = 100 + 0.8Y_D$, $Y_D = Y - T$, $I = 150$, $G = 200$, $T = 100$.

(a) **AE function.**

$$Y_D = Y - 100 \Rightarrow C = 100 + 0.8(Y - 100) = 20 + 0.8Y.$$

Therefore,

$$AE = C + I + G = (20 + 0.8Y) + 150 + 200 = \mathbf{370 + 0.8Y}.$$

(b) **Equilibrium Y .**

Set $Y = AE$:

$$Y = 370 + 0.8Y \Rightarrow 0.2Y = 370 \Rightarrow \boxed{Y = 1850}.$$

(c) **MPC.** $\boxed{\text{MPC} = 0.8.}$

(d) **Multiplier.** $\boxed{\text{Multiplier} = \frac{1}{1 - \text{MPC}} = \frac{1}{0.2} = 5.}$

(e) **Effect of $\Delta G = +50$.**

$$\Delta Y = \text{multiplier} \times \Delta G = 5 \times 50 = 250.$$

$$\text{New equilibrium: } \boxed{Y' = 1850 + 250 = 2100.}$$

7. Solution.

- (a) If actual output $>$ planned AE , sales fall short of production $\Rightarrow$ *unplanned inventory accumulation*. Firms respond by cutting production (and possibly prices), moving Y toward equilibrium.
- (b) If actual output $<$ planned AE , sales exceed production $\Rightarrow$ *unplanned inventory depletion*. Firms respond by raising production (and possibly prices), moving Y toward equilibrium.

8. *Given:* $NX = 50 - 0.1Y$, and parameters from Question 6.

Solution. With $C = 20 + 0.8Y$, $I = 150$, $G = 200$, and $NX = 50 - 0.1Y$, we have:

$$AE = (20 + 0.8Y) + 150 + 200 + (50 - 0.1Y) = \mathbf{420 + 0.7Y}.$$

Equilibrium:

$$Y = 420 + 0.7Y \Rightarrow 0.3Y = 420 \Rightarrow \boxed{Y = 1400}.$$

Multiplier intuition: The open-economy marginal propensity to import (MPM= 0.1) is a leakage that lowers the slope of the AE line (from 0.8 to 0.7 here), thus reducing the multiplier from $\frac{1}{1-0.8} = 5$ to $\frac{1}{1-0.7} = \frac{1}{0.3} \approx 3.\bar{3}$. Hence, income changes are dampened when imports rise with income.